CS3500+CS3510 — Operating Systems (+ lab)

Tier	A
Phase (order)	4
Do this first	CS2705, CS2600, CS2222 (C)
Estimated effort	14 weeks, ~9 h/week
Unlocks	CS4100, CS6847, CS6440, CS6868

Roadmap

1. Processes, threads, scheduling.
2. Synchronisation: locks, semaphores, condition variables, deadlocks.
3. Memory: paging, virtual memory, page replacement.
4. File systems, I/O, and (in the lab) system calls.
5. Read and modify xv6 along the way.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Introduction to Operating Systems | IIT Madras (39 videos). NPTEL *Introduction to Operating Systems* (Chester Rebeiro, IIT Madras): xv6-based, this is the IITM OS course.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Operating Systems - Prof. Chester (50 videos). IIT Madras BS *Operating Systems* (Chester Rebeiro): newer recording of the same teacher.
- CS 537: OS | University of Wisconsin-Madison (26 videos). Wisconsin CS 537 (OSTEP-based) — pairs with the free OSTEP book.
- UC Berkeley CS162 Operating Systems and System Programming, Fall 2020 (27 videos). Berkeley CS162 Fall 2020 (re-upload): deeper on synchronisation and file systems.
- MIT 6.S081 (21 videos). MIT 6.S081 xv6 lectures (re-upload): do the labs at pdos.csail.mit.edu/6.S081.

⚡ Short videos for intuition and revision

- Operating System (77 videos). Neso Academy *Operating System*: short, exam-style revision.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–4, 12	What an OS does, PC hardware, programs -> processes, booting. Set up xv6 (MIT 6.1810 site) and boot it in QEMU.
2	6–11	CPU sharing, memory management, virtual memory, MMU, segmentation, xv6 memory. Read xv6 book ch. 3; trace a page-table walk.
3	13–19	Processes, fork/exec/exit, system calls, interrupts, context switch. Add a new system call to xv6.
4	20–24	Scheduling: priority, multiprocessor, Linux CFS. Implement a priority or lottery scheduler in xv6.
5-6	25–34	IPC, synchronisation, Peterson/Bakery, hardware locks, mutexes, semaphores, dining philosophers, deadlocks. Solve producer-consumer and readers-writers with pthreads.
7	35–36	Threads. Write a user-level threads library (the xv6 *uthread* lab).
8	37–39, 5	OS security, information flow, buffer overflows. Then file systems from OSTEP ch. 39-42 (not in this playlist).

▼ **All lectures in *Introduction to Operating Systems* | *IIT Madras* (39 videos)**

1. #1 Introduction to the Course | Introduction to Operating Systems
2. #2 Introduction to Operating Systems
3. #3 PC Hardware | Introduction to Operating Systems
4. #4 From Programs to Processes | Introduction to Operating Systems
5. #38 Buffer Overflows | Introduction to Operating Systems
6. #5 Sharing the CPU | Introduction to Operating Systems
7. #6 Memory Management Introduction | Introduction to Operating Systems
8. #7 Virtual Memory | Introduction to Operating Systems
9. #8 MMU Mapping | Introduction to Operating Systems
10. #9 Segmentation | Introduction to Operating Systems
11. #10 Memory Management in xv6 | Introduction to Operating Systems
12. #11 PC Booting | Introduction to Operating Systems
13. #12 Introduction to Processes | Introduction to Operating Systems
14. #13 Create, Execute & Exit from a Process | Introduction to Operating Systems
15. #14 System Calls for Process Management | Introduction to Operating Systems
16. #15 Interrupts | Introduction to Operating Systems
17. #16 Interrupt Handling | Introduction to Operating Systems
18. #17 Software Interrupts & System Calls | Introduction to Operating Systems
19. #18 CPU Context Switching | Introduction to Operating Systems
20. #19 CPU Scheduling | Introduction to Operating Systems
21. #20 Priority Based Scheduling Algorithms | Introduction to Operating Systems
22. #21 Multi-Processor Scheduling | Introduction to Operating Systems

23. #22 Scheduling in Linux | Introduction to Operating Systems
24. #23 Completely Fair Scheduling | Introduction to Operating Systems
25. #24 Inter Process Communication | Introduction to Operating Systems
26. #25 Synchronization | Introduction to Operating Systems
27. #26 Software Solutions for Critical Sections | Introduction to Operating Systems
28. #27 Bakery Algorithm | Introduction to Operating Systems
29. #28 Hardware Locks | Introduction to Operating Systems
30. #29 Mutexes | Introduction to Operating Systems
31. #30 Semaphores | Introduction to Operating Systems
32. #31 Dining Philosophers Problem | Introduction to Operating Systems
33. #32 Deadlocks | Introduction to Operating Systems
34. #33 Dealing with Deadlocks | Introduction to Operating Systems
35. #34 Threads | Part 1 | Introduction to Operating Systems
36. #35 Threads | Part 2 | Introduction to Operating Systems
37. #36 Operating System Security | Introduction to Operating Systems
38. #37 Information Flow Policies | Introduction to Operating Systems
39. #39 Preventing Buffer Overflow Attacks | Introduction to Operating Systems

Books, notes and websites

- ✅ NPTEL *Introduction to Operating Systems* (Chester Rebeiro, IIT Madras) — nptel.ac.in/courses/106106144 (uses xv6)
- *Operating Systems: Three Easy Pieces* — ostep.org (free)
- MIT 6.S081 (xv6) — pdos.csail.mit.edu/6.S081
- Book: Tanenbaum, *Modern Operating Systems* (the course text)

Practice / proof of understanding

Do the OSTEP projects or xv6 labs (add a syscall, implement a scheduler, copy-on-write fork). Write a tiny shell in C.

What your IITM course sheet says

From the PDF: CS3500

Official description: The course introduces operations systems (OS) to under graduates. The students would be able to understand the working of an operating system, how it interfaces with computer hardware, and the various interfaces which software can use. Additionally students will learn to appreciate design aspects of the OS.

Official textbooks: 1. Modern Operating Systems- Andrew S Tanenbaum, Prentice Hall2. Design of the Unix Operating System - Maurice Bach, Prentice Hall.

Official references: 1. Operating System Concepts - Operating System Concepts, Sixth Edition, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, John Wiley & Sons Inc.2. Operating Systems - Principles and

Practice : Thomas Anderson and Michael Dahlin, Recursive Books, 2014.

From the PDF: CS3510

Official description: 1. Implementation of a command shell. 2. Implementation of Synchronization constructs. 3. Study of scheduling algorithms 4. Implementation of File system constructs 5. Advanced file system constructs Currently, the whole set of five experiments are done in Java, on top of a java minikernel. A Nachos based solution may also be used. For parts (4) and (5) Unix file system is used as a case study.

Official textbooks: NIL